Dr. Priya N. Raghunathan

Associate Professor of Computational Biology

Pittsburgh, PA | +1 (412) 555-0187 | p.raghunathan@example.edu | github.com/raghunathan-lab-example | raghunathanlab.example.edu

RESEARCH INTERESTS

Statistical and machine-learning methods for single-cell and spatial transcriptomics; probabilistic models of gene regulatory networks; reproducible computational pipelines for clinical genomics.

EDUCATION

University of California, Berkeley | Ph.D. in Computational Biology — 2008 - 2013

Dissertation: Hierarchical Bayesian models of transcriptional noise. Advisor: Prof. M. Lindqvist.

University of Michigan | M.S. in Statistics — 2006 - 2008

Indian Institute of Technology Madras | B.Tech. in Biotechnology — 2002 - 2006

ACADEMIC APPOINTMENTS

Carnegie Mellon University, Computational Biology Department | Associate Professor (with tenure) — Pittsburgh, PA | Jul 2021 - Present

- Lead a group of 7 Ph.D. students, 2 postdocs and 3 research staff.
- Courtesy appointment in Machine Learning Department (2022 - present).

Carnegie Mellon University, Computational Biology Department | Assistant Professor — Pittsburgh, PA | Jul 2016 - Jun 2021

- Established the lab; graduated 3 Ph.D. students.

Broad Institute of MIT and Harvard | Postdoctoral Fellow — Cambridge, MA | Sep 2013 - Jun 2016

- Single-cell methods group; Jane Coffin Childs Fellow.

JOURNAL ARTICLES

Spatially resolved gene regulatory networks from sparse transcriptomic maps | Raghunathan P.N.*, Okoye T., Brandt L., Feld S. Nature Methods 22(4):511-523 — 2025

Calibrated uncertainty for cell-type deconvolution | Okoye T., Liang Y., Raghunathan P.N. Genome Biology 25:188 — 2024

scNoise: a hierarchical model of transcriptional bursting at single-cell resolution | Liang Y., Raghunathan P.N. Cell Systems 14(2):97-110 — 2023

Benchmarking spatial clustering methods across 42 tissue atlases | Brandt L., Herrera A., ..., Raghunathan P.N. Nature Communications 13:6120 — 2022

Reproducible clinical genomics pipelines with provenance tracking | Feld S., Raghunathan P.N. Bioinformatics 37(19):3311-3318 — 2021

Variational inference for allele-specific expression in tumors | Herrera A., Kim J., Raghunathan P.N. PLOS Computational Biology 16(8):e1008021 — 2020

Gene-gene covariance from droplet-based scRNA-seq | Raghunathan P.N., Lindqvist M. Annals of Applied Statistics 13(3):1601-1630 — 2019

Single-cell atlas of the developing human thymus | Thymus Atlas Consortium (incl. Raghunathan P.N.). Science 367(6480):eaay3224 — 2018

Noise-aware differential expression for low-input RNA-seq | Raghunathan P.N., Weiss D., Regev-Haddad K. Genome Research 27(9):1525-1536 2017

Hierarchical Bayesian models of transcriptional noise | Raghunathan P.N., Lindqvist M. Journal of the American Statistical Association 110(512):1406-1419 2015

CONFERENCE PROCEEDINGS

Okoye T., Raghunathan P.N. Optimal transport for spatial alignment of serial sections. ICML 2024, pp. 38112-38129., Liang Y., Okoye T., Raghunathan P.N. Contrastive pretraining for gene programs. RECOMB 2023, LNCS 13976:201-216., Brandt L., Raghunathan P.N. Graph neural networks for tissue niches. ISMB 2022 (Bioinformatics 38 Suppl 1)., Herrera A., Raghunathan P.N. Amortized inference for clonal deconvolution. NeurIPS 2020, pp. 11843-11855., Raghunathan P.N. Sparse factor models for single-cell multi-omics. ISMB 2017 (Outstanding Paper Award)., Raghunathan P.N., Weiss D. Bursting kinetics from snapshot data. RECOMB 2016, LNCS 9649:254-268.

GRANTS & FUNDING

R01 GM148812: Probabilistic models of spatial gene regulation | National Institutes of Health (NIGMS). PI. $2,450,000 2023 - 2028

CAREER: Uncertainty-aware machine learning for single-cell genomics | National Science Foundation (DBI-1945xxx). PI. $812,000 2019 - 2025

Chan Zuckerberg Initiative Single-Cell Biology Data Insights | CZI. Co-PI (with S. Feld). $400,000 2022 - 2024

Sloan Research Fellowship | Alfred P. Sloan Foundation. PI. $75,000 2018 - 2020

Pilot: Clinical-grade pipelines for pediatric tumor sequencing | UPMC Hillman Cancer Center. Co-I. $150,000 2017 - 2018

TEACHING

02-710 Computational Genomics (graduate) | Carnegie Mellon University. Instructor, ~60 students/yr. FCE 4.7/5 2017 - Present

02-251 Great Ideas in Computational Biology (undergraduate) | Carnegie Mellon University. Instructor, ~120 students/yr 2019 - Present

02-750 Automation of Scientific Research (graduate seminar) | Carnegie Mellon University. Co-instructor 2021 - 2023

Workshop: Bayesian Methods for Single-Cell Data | Cold Spring Harbor Laboratory course. Lecturer 2018, 2022

INVITED TALKS

Keynote: Uncertainty is a feature, not a bug, in spatial omics | Single Cell Genomics Conference, Stockholm 2025

Spatial gene regulation from sparse maps | Department Seminar, Stanford University 2024

Calibrated deconvolution for clinical samples | Broad Institute Models, Inference & Algorithms 2024

Probabilistic programming for genomics | Gordon Research Conference: Quantitative Genetics 2023

Transcriptional bursting at scale | EMBL Heidelberg Seminar Series 2022

Benchmarks that matter | Human Cell Atlas General Meeting 2022

Noise models for single-cell data | Joint Statistical Meetings, invited session 2019

Reproducibility in computational biology | NIH Data Science Symposium 2018

PROFESSIONAL SERVICE

Associate Editor, PLOS Computational Biology (2021 - present), Program Committee, ISMB (2019 - 2025), RECOMB (2021 - 2024), NeurIPS (2020 - 2023), NIH Study Section ad hoc reviewer, GCAT and MABS (2020 - 2024), Reviewer: Nature Methods, Genome Biology, Cell Systems, Bioinformatics, JASA, Chair, CMU Computational Biology Ph.D. Admissions (2022 - 2024), Co-organizer, Pittsburgh Single-Cell Symposium (2019, 2023)

HONORS & SKILLS

Technical Skills: R, Python, Stan, PyTorch, Nextflow, Slurm/HPC, scRNA-seq, Spatial transcriptomics

Languages: English, Tamil, Hindi

Awards: NSF CAREER Award (2019), Sloan Research Fellowship in Computational & Evolutionary Molecular Biology (2018), CMU School of Computer Science Teaching Award (2020), ISMB Outstanding Paper Award (2017), Jane Coffin Childs Postdoctoral Fellowship (2013 - 2016)